

MEC 305 — Thermofluids Engineering

Faculty of Engineering and Architectural Science · Department of Mechanical and Industrial Engineering · Fall 2026 · 4.0 credits · Sections 01–03

COURSE OUTLINE AND STUDENT INFORMATION PACKAGE

This document constitutes the official course outline for MEC 305 for the Fall 2026 term. Students are responsible for the entire contents of this document. Enrolment in this course is taken as acknowledgement that the student has read and understood all sections, including the sections concerning laboratory safety, academic integrity, and the mandatory field study.

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1. Instructional Team

Role	Name	Contact	Office hours
Course instructor	Dr. Priya Raghunathan, P.Eng.	praghunathan@example.edu	Tue 13:00–15:00, ENG 429
Section 02 instructor	Dr. Tomasz Wierzbicki	twierzbicki@example.edu	Wed 10:00–12:00, ENG 431
Laboratory coordinator	M. Ferreira, M.A.Sc.	mferreira@example.edu	Thu 14:00–16:00, ENG 108
Head teaching assistant	S. Adeyemi	sadeyemi@example.edu	By appointment

2. Course Description

MEC 305 integrates the principles of thermodynamics, fluid mechanics, and heat transfer into a unified treatment of energy systems. Topics include conservation of mass, momentum and energy in control volumes; the first and second laws applied to open systems; internal and external viscous flow; dimensional analysis and similitude; conduction, convection and radiation heat transfer; and the analysis of heat exchangers. The course culminates in an open-ended design study in which student teams size and specify a thermal system against a realistic performance and cost brief.

Prerequisites: MEC 222 (Thermodynamics I) and MEC 231 (Fluid Mechanics I), or equivalent. Students who have not completed both prerequisites will be removed from the course roster without notice during the second week of term.

3. Learning Outcomes

On successful completion of this course, students will be able to:

3.1 Formulate and apply conservation statements for mass, momentum and energy to open thermodynamic systems, identifying appropriate control volumes and justifying all simplifying assumptions.

3.2 Evaluate the thermodynamic performance of power and refrigeration cycles, including the calculation of thermal efficiency, coefficient of performance, and second-law efficiency.

3.3 Characterize internal and external viscous flows, computing friction losses, drag, and pressure drop across piping networks and immersed bodies.

3.4 Apply dimensional analysis and the Buckingham Pi theorem to reduce experimental parameter spaces and to design physically meaningful model tests.

3.5 Analyze steady and transient conduction in one and two dimensions using analytical and finite-difference methods, and assess the validity of the lumped capacitance approximation.

3.6 Select and apply appropriate empirical correlations for forced and natural convection, and quantify the uncertainty introduced by correlation choice.

3.7 Size and rate heat exchangers using both the log-mean temperature difference and effectiveness–NTU methods, and justify the selection of one method over the other.

3.8 Work effectively in a multidisciplinary team to deliver an engineering design study against a written brief, including the preparation of professional-quality technical documentation.

3.9 Communicate quantitative engineering results in written and oral form to both technical and non-technical audiences.

4. Graduate Attributes

This course contributes to the following CEAB graduate attributes, which are assessed through the indicated instruments:

Attribute	Level	Assessed by
Knowledge base for engineering	Developed	Midterm, final examination
Problem analysis	Developed	Assignments, midterm
Investigation	Introduced	Laboratory program
Design	Developed	Design study report
Use of engineering tools	Developed	Laboratory, design study
Individual and team work	Developed	Design study
Communication skills	Developed	Design study report and presentation
Professionalism	Introduced	Field study, laboratory conduct
Impact on society and environment	Introduced	Design study
Ethics and equity	Introduced	Academic integrity module
Economics and project management	Introduced	Design study cost brief
Life-long learning	Introduced	Reflective memorandum

5. Delivery and Weekly Schedule

Week	Beginning	Lecture topic	Laboratory	Due
1	Sep 8	Control volume formulation; review	No lab	—
2	Sep 14	First law for open systems	Safety orientation (mandatory)	—
3	Sep 21	Second law; entropy generation	Lab 1 — Pipe friction	Assignment 1
4	Sep 28	Power cycles: Rankine	Lab 1 continued	Field study deposit
5	Oct 5	Refrigeration and heat pump cycles	Lab 2 — Venturi and orifice	Assignment 2
—	Oct 12	Fall reading week — no classes or labs	—	—
6	Oct 19	Internal viscous flow; Moody chart	Lab 2 continued	—
7	Oct 26	External flow; boundary layers	No lab (midterm week)	Midterm Oct 28
8	Nov 2	Dimensional analysis and similitude	Lab 3 — Drag on bluff bodies	Assignment 3
9	Nov 9	Steady conduction; fins	Lab 3 continued	Field study balance
10	Nov 16	Transient conduction	Lab 4 — Transient cooling	Assignment 4
11	Nov 23	Forced and natural convection	Lab 4 continued	—
12	Nov 30	Heat exchangers; LMTD and ϵ -NTU	Design consultations	Design report Nov 30
13	Dec 7	Radiation; course synthesis	No lab	Reflective memo Dec 4

6. Laboratory Program

The laboratory program consists of four two-week experiments plus a mandatory safety orientation. Laboratory attendance is compulsory. A student who misses the safety orientation may not enter the laboratory for any subsequent session and will therefore fail the laboratory component, which is a barrier requirement for the course.

6.1 Laboratory groups are assigned by the laboratory coordinator in week one and may not be changed.

6.2 Pre-laboratory preparation must be submitted electronically before 09:00 on the day of the session. Students who have not submitted pre-laboratory work will not be admitted.

6.3 Laboratory reports are due ten calendar days after the second session of each experiment.

6.4 Personal protective equipment is mandatory in all laboratory spaces. See section 13.

6.5 Data falsification in a laboratory report is treated as academic misconduct under section 11 and is reported to the Academic Integrity Office without exception.

6.6 Laboratory Conduct in Detail

Students shall arrive at the laboratory not less than five minutes before the scheduled start time. The laboratory door is locked at the scheduled start time and late students are recorded as absent for the session. There is no appeal from this rule; it exists because the safety briefing that opens each session cannot be repeated for individual latecomers.

All apparatus shall be returned to its designated storage position at the end of each session. Groups whose bench is left disordered forfeit five percent of the report grade for that experiment. The laboratory coordinator's assessment of bench condition is final.

Data acquisition is performed using the departmental LabVIEW installation. Students may not install additional software on laboratory machines, may not connect personal storage devices to laboratory machines, and shall transfer data using the departmental network share provided for that purpose.

Where an experiment produces anomalous results, students shall report the anomaly in the discussion section of the report rather than discarding the data. Marks are awarded for the quality of the analysis, not for the closeness of the result to a textbook value. Reports that silently omit inconvenient data points are referred under section 11.

Groups requiring additional laboratory time may request supervised access during the coordinator's posted hours. Unsupervised access to the laboratory is prohibited at all times and is a dismissible offence for students holding a departmental key.

The department maintains a calibration log for all instruments. Students shall record the instrument serial number and last calibration date in every report. Reports that omit calibration information are returned unmarked and are subject to the late penalty from the date of return.

6.7 Report Formatting Requirements

Reports shall be prepared in the departmental template, which is available from the course portal. The template specifies 11-point Arial, single spacing, 25 mm margins, and numbered section headings. Figures shall be numbered consecutively and referenced in the text before they appear. Tables shall be numbered separately from figures. Every figure and table requires a caption; figure captions appear below the figure and table captions above the table. Units shall be SI throughout, with the exception of pressures, which may be reported in bar where this is conventional for the apparatus in question. Uncertainty shall be propagated and reported for every derived quantity; a derived quantity reported without an uncertainty estimate receives no marks regardless of its numerical accuracy.

Reports exceeding the page limit stated in the experiment brief are truncated at the limit and marked on the truncated content only. Appendices do not count toward the page limit but are not marked; do not place substantive analysis in an appendix and expect it to be read.

6.8 Equipment Damage and Replacement

Students are not held financially liable for equipment failures arising from ordinary use. Damage arising from a failure to follow the written procedure, from unauthorized modification of apparatus, or from horseplay is charged to the responsible student at replacement cost. Such charges are applied to the student account and are separate from the deposits and fees described in section 7 of this outline.

The department's insurer requires that all incidents involving apparatus damage be documented within twenty-four hours. Students shall report damage immediately to the laboratory coordinator, whether or not they believe themselves responsible. Concealment of damage is treated as a professional conduct matter under section 14.

7. Required Resources and Student Costs

The following costs are associated with participation in MEC 305. Students who anticipate difficulty meeting these costs should contact the Faculty's Student Affairs office before the end of week two; a limited hardship fund exists and applications are confidential.

Item	Cost	Payable to	Due date
Laboratory kit (multimeter, thermocouple set, safety glasses, log book)	\$88.00	Campus engineering store	September 14, 2026
Textbook: Fundamentals of Thermal-Fluid Sciences, 6th edition	\$232.00 new \$168.00 used	Campus bookstore	September 8, 2026
Solutions supplement (optional)	\$41.00	Campus bookstore	Optional
MATLAB and Simulink student licence	\$0.00 — provided by the university	—	—
Field study deposit (see section 8)	\$150.00	Department, via portal	September 30, 2026
Field study balance (see section 8)	\$175.00	Department, via portal	October 24, 2026
Departmental locker deposit (refundable)	\$20.00	Department office	September 14, 2026

The total non-optional cost of participation, assuming a used textbook, is **\$601.00**, of which **\$20.00** is refunded at the end of term.

8. Mandatory Field Study

MEC 305 includes a mandatory two-day field study to an operating combined-cycle generating station and district energy plant. The field study takes place on **November 5 and 6, 2026** and replaces lectures and laboratories on those dates. Attendance is a barrier requirement: a student who does not attend cannot pass the course, and the field study cannot be completed remotely or in a subsequent term.

The total cost of the field study is \$325.00 per student, comprising coach transport, two nights' shared accommodation, site insurance, and required personal protective equipment issued on site. This is collected in two instalments: a non-refundable deposit of **\$150.00 due September 30, 2026**, and the balance of **\$175.00 due October 24, 2026**. Students who have not paid the balance by the due date are removed from the manifest and, per the preceding paragraph, cannot pass the course.

Meals during the field study are not included and students should budget approximately \$60.00 to \$90.00 for meals across the two days. One group dinner is provided by the host utility.

9. Evaluation

Component	Weight	Date	Barrier?
Assignments (4)	8%	See section 5	No
Laboratory program (4 experiments)	20%	Reports due +10 days	Yes — 50% required
Midterm examination	22%	Wednesday, October 28, 2026	No
Design study report	15%	Monday, November 30, 2026	No
Reflective memorandum	2%	Friday, December 4, 2026	No
Final examination	33%	Wednesday, December 16, 2026	Yes — 40% required

A student who does not achieve the barrier minima in the laboratory program and the final examination receives a maximum grade of D- regardless of the weighted total.

10. Examination Policy

10.1 The midterm examination is held on **Wednesday, October 28, 2026**, 18:00–20:00. Room assignments are posted to the portal one week in advance and are by surname.

10.2 The final examination is scheduled by the Registrar for **Wednesday, December 16, 2026**. Students shall not make travel arrangements that conflict with the examination period, which runs from December 9 to December 18, 2026.

10.3 One double-sided aid sheet, handwritten, is permitted in both examinations. Photocopied or printed aid sheets are confiscated at the start of the examination.

10.4 Only the departmentally approved calculator model may be used. A list of approved models is posted to the portal. Programmable and graphing calculators are not approved.

10.5 A missed midterm examination with documented cause has its weight transferred to the final examination. There is no deferred midterm sitting.

10.6 A missed final examination is administered by the Registrar's deferred examination process. Application is made to the Registrar, not to the instructor.

11. Academic Integrity

The University's Policy 60 on Academic Integrity applies in full to this course. Students are expected to have read it. The following clarifications are specific to MEC 305 and do not narrow the policy in any way.

11.1 Assignments may be discussed with classmates. Written solutions must be prepared independently. Submitting a solution substantially identical to another student's is an offence for both parties.

11.2 Laboratory data is collected as a group and may be shared within the group only. Analysis and discussion are individual work.

11.3 Generative artificial intelligence tools may be used for language editing and for exploring concepts. They may not be used to generate analysis, calculations, figures, or discussion that is submitted as the student's own work. Where such tools are used for editing, the use must be disclosed in the acknowledgements.

11.4 Fabrication or falsification of experimental data is among the most serious offences in engineering practice and is treated accordingly. There is no minor instance of data falsification.

11.5 The reuse of a report or portion of a report submitted for credit in another course requires the written permission of both instructors in advance.

12. Academic Accommodation

Students with disabilities are encouraged to register with Academic Accommodation Support. Accommodation is arranged through that office and is implemented by the instructor on receipt of an accommodation letter. Students should register as early in the term as possible; accommodations are not applied retroactively to assessments already completed. Requests for examination accommodation must reach the office by the deadlines it publishes, which for the December examination period fall in early November.

Students requiring accommodation on religious, Aboriginal or spiritual grounds shall submit the relevant form within the first two weeks of the term, or as soon as the need is known where the date falls outside that window.

13. Health and Safety

Personal protective equipment consisting of CSA-approved safety glasses, closed-toe footwear and, where posted, hearing protection is mandatory in all laboratory and shop spaces. Safety glasses are included in the laboratory kit described in section 7. Long hair shall be tied back and loose clothing secured when working near rotating apparatus. Students shall complete the online WHMIS and laboratory safety modules before the week two orientation; the modules take approximately ninety minutes and are free of charge.

The field study site requires additional protective equipment which is issued on arrival and is included in the field study cost. Students who arrive at the site without closed-toe footwear and long trousers are refused entry and, as the field study is a barrier requirement, will be unable to complete the course.

14. Professional Conduct

Students in this course are preparing to enter a self-regulating profession. Conduct in laboratories, on the field study, and in correspondence with the instructional team is expected to meet the standard that would be expected of a candidate member of a provincial engineering association. Discourtesy toward technical staff, host-site personnel, or fellow students is a matter for the Faculty's professional conduct process, which operates independently of the academic integrity process.

Correspondence shall be sent from the student's university address and shall identify the course and section. The instructional team undertakes to reply to correspondence within two business days during term. Correspondence sent in the forty-eight hours before an assessment may not receive a reply before that assessment.

15. Important Dates

Date	Event
September 8, 2026	Classes begin
September 14, 2026	Laboratory kit and locker deposit due; safety orientation week
September 21, 2026	Last day to add a course; last day to drop with 100% refund
September 30, 2026	Field study deposit due — \$150.00
October 2, 2026	Last day to drop with 50% refund
October 12–16, 2026	Fall reading week — no classes
October 16, 2026	Last day for any tuition refund on dropped courses
October 24, 2026	Field study balance due — \$175.00
October 28, 2026	Midterm examination, 18:00–20:00
November 5–6, 2026	Mandatory field study — no lectures or laboratories
November 10, 2026	Last day to drop without academic penalty
November 30, 2026	Design study report due
December 4, 2026	Reflective memorandum due
December 7, 2026	Last day of classes
December 9–18, 2026	Final examination period
December 16, 2026	MEC 305 final examination

This outline was approved by the Department Curriculum Committee on August 14, 2026. In the event of a discrepancy between this outline and the University Calendar, the Calendar governs. The instructor reserves the right to amend the schedule of topics; assessment weights and dates will not be changed after the second week of term except in the case of a University closure.